



João Pedro Dudziak Fonseca

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EDUCATION

Georgia Institute of Technology, Atlanta, GA

B.S. in Mechanical Engineering, Minor in Computer Science (Computing & Intelligence)

Graduation: Dec 2026

GPA: 3.89/4.00

EXPERIENCE

Vehicle Dynamics & Modeling Engineering Intern | Tesla, Palo Alto, CA

Jan 2026 - Jun 2026

- Designed suspension spring pads and strut-tower braces in CATIA; validated with K&C testing, FEA/hand calculations, prototyping, and DFMA; achieved the target 10× rear lateral stiffness increase for a vehicle program.
- Developed test procedures and data analyses end-to-end for Stepsteer and Sinusoid maneuvers, including ISO-compliance, maneuver extraction, data processing, metrics, plots and eLSD performance/vehicle dynamics insights.
- Rebuilt all I/O infrastructure of Data Analysis Tool, contributing 20k+ LOC (130+ files), for MoTeC/Modelica parsing, ISO filtering/delay, unit conversion, sign/location transformations, and DataFrame validation with Pydantic.
- Developed a Python shape-constrained spline-fitting library that auto-detects data landmarks (minima/maxima/etc.) to apply physics-based constraints without per-trial retuning; integrated into all maneuver-analysis pipelines.
- Built ImGui app from scratch (20k LOC) for all multibody-dynamics (MBD) simulation workflows, including Vehicle/Controller YAML editing, FMU generation, MicroSim execution, MongoDB upload, DAT metric/plot integration.

Underwater Jet Propulsion Researcher | Bhamla Lab (DARPA Funded), Atlanta, GA

May 2025 - Dec 2025

- Conducted experimental/computational research on squid-inspired compliant nozzles; designed experimental methods and collected chromatophore and SLEAP-based squid kinematics data used in the [2026 arXiv preprint](#).
- Developed transparent, tracer-embedded 3D printed flexible nozzles and pulsed-jet laser-based experiments using 3D PTV, and high-speed imaging for synchronized nozzle deformation and flow visualization.
- Validated Fluid-Structure Interaction (FSI) simulation models by comparing experimental flow to numerical predictions, contributing to the development of a Physics-Informed Neural Network (PINN) optimization framework for nozzle shape.

Equipment Engineering Intern | Coca-Cola (2 Rotations), Atlanta, GA

May 2023 - Aug 2023, Jan 2024 - May 2024

- Designed and printed custom Gas Volume Testers for dispensed liquid Carbonation Test (CT), automating current methods by using a vibrating table. Cut test-time in half; improved measurement resolution by 2 decimal places.
- Researched and integrated palm recognition payment in vending machines, estimated to cut customer journey time by 10%. Secured a partnership for the technology, kickstarting a company worldwide project, now in market trials.
- Analyzed tipping angles for solar-powered vending machines with FEA/hand-calculations, validating FoS and ISO standards.

Aerodynamics & Chassis Engineer | GT FSAE HyTech Racing, Atlanta, GA

Sep 2022 - Dec 2025

- Designed suspension wishbones for the car in CATIA; validated aerodynamic behavior with Fluent and wind tunnel testing.
- Adjusted the front-wing endplate's angle of attack by 3 degrees based on F1 aerodynamic literature, increasing downforce by 4 kg at 120 km/h with negligible drag penalty, improving stability in high-speed cornering.

Production Engineering Intern | Sorvetes Rochinha Ice Cream CO, São Paulo, Brazil

May 2022 - Aug 2022

- Applied process mapping to the entire production line and identified packaging as a bottleneck.
- Reduced packaging prep time by 1 hour (per 8 hour shift) by designing a mount for an automatic labelling machine.

PROJECTS

Founder & Lead Engineer | TrailMate (Follower Robot Backpack), Atlanta, GA, [Link](#)

Apr 2025 - Present

- Designed and built a follower robot capable of carrying 20+ kg, including powertrain (geared DC motor with 1:9 transmission), steering (servo-driven rack and pinion), suspension (double-wishbone) using CAD / FEA stress and buckling analysis in NX.
- Developed real-time control pipeline on RaspberryPi and STM32 with LiDAR SLAM mapping for obstacle avoidance and transmitter-receiver trilateration for user tracking, implementing a scoring algorithm to compute steering/throttle commands.

Inverse Kinematics Approximation for Robotic Arms with Machine Learning, Atlanta, GA, [Link](#)

May 2025 - Aug 2025

- Developed a machine learning framework for robotic arm inverse kinematics, achieving sub-degree joint accuracy and sub-millisecond inference for real-time, high-DOF manipulators; awarded "Best Project" among 30+ teams.
- Built supervised (MLP/Gradient Boosting), unsupervised (K-Means/DBSCAN) models with PyTorch/Scikit-Learn for reachability/joint-space regression on 1M+ samples; 10x lower error/10x faster inference over PyBullet IK baseline solver.

SKILLS & HONORS

UKMT Mathematics Challenge Gold Medalist - 2020 | 2021

3x Faculty Honors, 4x Dean's List - 2022-2026

Engineering: CATIA6, SolidWorks, Siemens NX, Ansys, Dymola, Simulink, CFD, FEA, 3D Printing, DFMA, BoMs, NVH, KiCad

Software: Python, C/C++, Matlab, Java, Modelica, PyTorch, Scikit-learn, NumPy, Data Pipelines, MongoDB, Express, React